Exercise 2: Sabre Dance

Difficulty Rating ★★☆☆☆

Document Team

October 30, 2026

The original concept for Sabre Dance was created by @snrkatat. Sound files were also made by him with the exception of Mystery01.wav. All of the exercises in this guide follow a similar approach.

1 Aims and Objectives

The aim of this exercise is to provide a practical example of how to use the LOFAR and DEMON screens to identify the acoustic signature of a new OPFOR frigate class. The exercise will also provide an opportunity to practice the use of the MACS database and vessel matching system. At the end of this exercise you should be able to:

1. To make the information given in Exercise 1 concrete by practical application.
2. Use the LOFAR and DEMON screens to identify the acoustic signature of a new OPFOR frigate class.
3. Assess changes in the display as the ship changes speed or course and identify the features that are speed dependent and those that are not.
4. Use the MACS database and vessel matching system to establish if a new recording is the same vessel.
5. To make reasonable assumptions about the ship's speed
6. To prepare a report that will allow future operators to quickly identify this class of ship.

2 Setup Instructions

1. Download the "Sabre Dance Package.zip" from the GitHub Repository.
2. Extract and copy the 6 sound files to your MACS recordings folder typically `\Documents\MACS\Recordings` on Windows but may vary if using OneDrive or the location of the recordings folder has been changed in Settings. If using Linux or MACos it is recommended to leave the recordings folder in the default location of `/home/<user>/Documents/MACS/Recordings`.

3. Start MACS and select LOFAR in Panel A and PSD in Panel B.
4. Initially set the LOFAR and PSD Palettes to Thermal and FFT to 131072 and in the LOFAR panel set f_{min} to 0 and f_{max} to 1000 and Normalisation (Norm) on.
5. Click the play button (green triangle) for the `sabre_flank_28kt` recording, set playback to loop then wait for 2-3 minutes while data accumulates.
6. Adjust the Floor and Dyn sliders to get the best image. Floor -27 and Dyn 123 works well.
7. You are now ready to start the exercise.
8. The remaining files are for use in the exercise.

3 The Mission

The Officer of the Deck has just finished briefing your sonar team. Earlier this morning a message received over the AN/BRA-24 caused much head scratching and debate between the Commander, Executive Officer and the OOD but now everyone is excited to know what's up.

A local asset has reported that a new OPFOR guided-missile frigate has departed its home port for a work-up cruise. Satellite has confirmed its sailing. The frigate is a new class, code name Sabre.

Some information is known about the class. It is rumoured to be a modern 2 propeller shaft diesel engine ship in a co-diesel and diesel (CODAD) 4×20 cylinder diesels (2 per shaft) configuration (see "Example Engine.pdf" in package) . It is also known to have fixed 4 bladed propellers with a diameter (ϕ) of 4.4 meters with a designed pitch to diameter (P/D) ratio of 1.1. With an estimated 6.0 turns per knot (TPK); there is no information about slippage. Satellite analysis suggests the frigate is about 134m in length with a beam of 16m. Displacement is estimated to be about 4,000 tons and sustained tracking has seen the ship routinely sailing between 14 and 28 knots.

You are tasked with intercepting and approaching the frigate in order to gain as much acoustic information as possible. You are to analyse this data and create a record in the ships sonar database so that ships of this class, encountered in the future, can be quickly identified.

3.1 Task 1 - Fundamental feature identification

Using: `sabre_cruise_14kt.wav` and `sabre_flank_28kt.wav`

Complete as much of the table provided in the package ("Blank Table.pdf" designed to be printed in landscape) for 14 kts and 28 kts as is possible. Hint it is possible to fill it completely with the information available and from the LOFAR / DEMON screens although some of the tonals can only be confirmed in Task 3. It may help to re-read the worked example at the beginning of the Illustrated Examples and Exercises.

Note: Items in the table may not be in the optimal solve order! It is recommended to print the table and complete it by hand while working through the exercise. Also, although this synthetic example demonstrates gearing and associated sidebands, be aware that gearing is not modelled in MNW yet so will not be apparent in game.

Note: In this exercise Cylinder Firing Rate (CFR) is used instead of the more correct Engine Firing Rate (EFR). CFR is not incorrect but risks confusion with Individual Cylinder Firing Rate (ICFR). Bear in mind that CFR $\equiv$ EFR.

For example (fake data), rule text may be a formula or free text. If harmonics are seen include them in brackets.

Feature	Rule	Cruise (14kts)	High Speed (28kts)
Shaft Rate (SR)	$RPM \div 60$	1.8 Hz	2.9 Hz
Mains (fixed)	50 Hz (+100, 150, 200)	50 Hz	50 Hz

3.2 Task 2 - Produce a finished report

Fill in the provided LOFARgram blank ("Blank LOFARgram.pdf" in package) and combine it with your completed table for the features you have identified in task 1.

3.3 Task 3 - Effect of speed and direction changes

Analyse in MACS the following sound files:

- `sabre_speed_change.wav`
- `sabre_turn_20kt.wav`

Then answer the following questions.

1. Why do some of the lines bend when the speed changes?
2. Why do some of the lines remain fixed as speed changes?
3. What is causing the lines to be wavy when the ship is turning (hint: this is a well known sound effect)?
4. Why are all the lines affected as the ship turns but not when speed changes?

3.4 Task 4 - Speed estimate

The purpose of this task is to examine the results from the LOFARgram and estimate the ship's speed. Since we already have estimates for both our recordings of 14 and 28 knots our task is to review the different ways we can calculate speed and confirm them against known reference points. This will give you confidence when you are required to determine speed in the future when no reference points are available.

Method 1: Turns Per Knot

1. Identify the correct shaft rate (SR in Hz) for the ship at 14 knots and 28 knots.
2. Calculate the RPM for each SR using the formula $RPM = SR \times 60$.
3. Calculate the speed using the formula $Speed = \frac{RPM}{TPK}$ where TPK is turns per knot.
4. Compare your calculated speeds with the known speeds of 14 and 28 knots.

Method 2: Propeller Pitch This method requires an understanding of the relationship between propeller pitch, shaft rate and speed. It is also more sensitive to errors in the pitch value. The pitch value is often not known accurately and can vary with propeller design, wear and fouling. However, it is still a useful method to cross-check your results.

1. Identify the correct shaft rate (SR in Hz) for the ship at 14 knots and 28 knots.
2. Calculate the RPM for each SR using the formula $RPM = SR \times 60$.
3. Calculate the propeller pitch P $1.1 * 4.5 = 4.95$ (P/D ratio $\times\phi$)
4. Calculate the speed using the formula $Speed = \frac{RPM \times 60 \times P}{1852}$ where P is propeller pitch in meters, 1852 is the number of meters in a nautical mile and the $\times 60$ converts RPM to revolutions per hour. **Note:** Sometimes the pitch may be given in feet or yards. It is left to the reader to find out the number of feet in a nautical mile. In every case a good submariner will be recording details like this in a personal notebook for future reference.
5. Compare your calculated speeds with the known speeds of 14 and 28 knots.

Points to Ponder: Is there a difference between the two methods? Which method is more reliable? Why might that be the case?

You receive additional information from the local asset. The manufacturer of the propellers has leaked that the propeller pitch is 4.95 meters and the turns per knot is 6.0. but the manufacturer has also confirmed that the propellers are designed to operate with a 15% slip. This means that the actual speed of the ship will be 15% less than the speed calculated using the propeller pitch method. Recalculate the speed using the propeller pitch method with the slip factored in and compare it with the known speeds of 14 and 28 knots. How does this affect your confidence in the propeller pitch method?

Hint: The slip factor is a percentage of the calculated speed. For example, if the calculated speed is 20 knots and the slip factor is 15%, then the actual speed would be $20 - (20 \times 0.15) = 17$ knots. For more details see Appendix B page 50 for an alternative method of calculating speed including slip and mixed units. Step By Step solution to this problem is in the EX2 Answers.pdf in the package.

3.5 Task 5 - Vessel Matching and Classification

The aim of this task is to demonstrate the process of using the cursor system to mark tonals on the LOFARgram and then adding the marked up vessel to the MACS database. Once our contact is in the database we will then use it to attempt to classify unknown vessels from new recordings.

Step 1: Mark up and store a contact in the database

1. Start the `sabre_cruise_14kt` recording in MACS, set playback to loop.
2. Set Panel A to Lofargram, $f_{min} = 0$ and $f_{max} = 300$ (Floor = -10 and Dyn = 127 with Normalisation on works for me but adjust to suit your own system.)
3. Start by assessing the clearly audible shaft and propeller noise (see EX1 for reminder on how to do this.)
4. Drill in to the low frequency range by reducing $f_{max} = 100$, you may need to adjust the Floor and Dyn to sharpen ths picture
5. You should be able to identify the shaft fundamental, blade rate, crankshaft rate, cylinder firing rate, gear mesh and electrical mains from earlier in the exercise.
6. Switch Panel B to Demon and confirm what tonals you can (Note the 8.4 hz crankshaft rate is clearly separated from the blade harmonics in DEMON).

A guide for using the harmonic and sideband cursors are left for another exercise.

7. Using the "+Classic" cursor mark the key tonals you have identified. Adjust f_{min} and f_{max} to zoom in and out to make this easier and cursor placement more accurate.
8. When the chosen tonals are selected, in the right panel new contact area enter a new contact name and an optional description. Click the "+ Create with marked frequencies" button.
9. This will open the "+ Add Measurement" page. Enter available additional information and make sure to save and close the add measurement screen.

Step 2: Add a new measurement to your new stored contact. In this step you have an oportunity to measure some tonals for the same ship but this time it is doing 28 knots.

1. Repeat steps 1 - 7 from the first step using `sabre_flank_28kt`. You will need to extend $f_{max} \approx 200$ to see relevant all tonals.
2. **Do not create a new contact** instead click "Browse all contacts ..." and scroll down to your previously added contact
3. Click the open folder icon to bring up the contact details and select the add measurement tab. Your new selected tonals will appear.
4. Add supporting detail, in particular make sure the estimated speed is entered.
5. Save and close

You now have two recordings for the same ship but with different parameters. Should you encounter this ship again you may be able to match your recorded values with the tonals from the new ship and so quickly identify it.

Step 2: Match a target using the database.

1. Play `Mystery01.wav` and do an audible analysis to get SR and BR; mark them with the cursor.
2. Explore the LOFAR and DEMON and note and mark any obvious tonals. **Hint:** Extend the f_{max} to 1500 hz. What do you think is going on up there.
3. Once you have marked any candidate tonals click **Compare marks against database** in top right panel.
4. The **Comparison results** panel opens. Select Fail or Partial or Pass (below the Filter By Name text box) and view the results.
5. The tolerance of the matching algorithm can be tuned in th **Target known details** panel. You can also enter known details such as SR and CSR to help pinpoint the ID.

Is Mystery01 Sabre?

Repeat Step 2 but this time use `Mystery02.wav` does this match Sabre?

The matching system in MACS allows for a great deal of fine tuning which is not demonstrated in this example. It is highly recommended that you spend time exploring all the features using the sound files from this exercise.